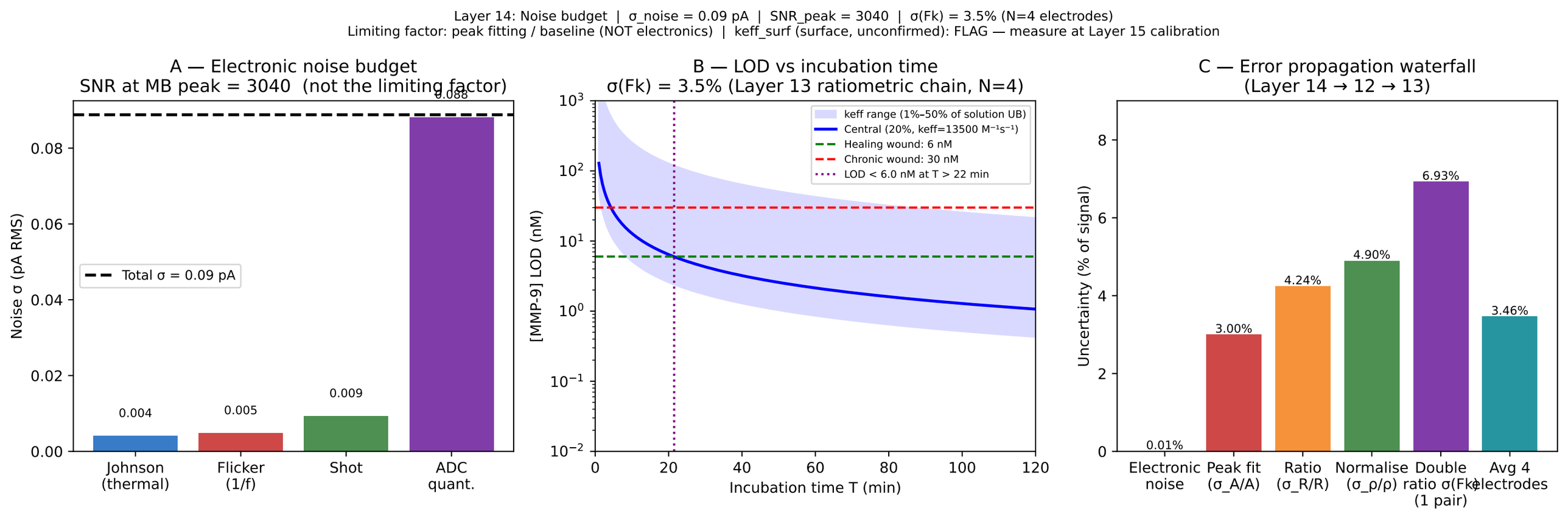
Layer 14: Noise budget | σ_noise = 0.09 pA | SNR_peak = 3040 | σ(Fk) = 3.5% (N=4 electrodes)
Limiting factor: peak fitting / baseline (NOT electronics) | keff_surf (surface, unconfirmed): FLAG — measure at Layer 15 calibration
A — Electronic noise budget
SNR at MB peak = 3040 (not the limiting factor)
Noise σ (pA RMS)
Total σ = 0.09 pA
0.004
0.005
0.009
0.088
Johnson (thermal)
Flicker (1/f)
Shot
ADC quant.
B — LOD vs incubation time
σ(Fk) = 3.5% (Layer 13 ratiometric chain, N=4)
[MMP-9] LOD (nM)
keff range (1%–50% of solution UB)
Central (20%, keff=13500 M⁻¹s⁻¹)
Healing wound: 6 nM
Chronic wound: 30 nM
LOD < 6.0 nM at T > 22 min
Incubation time T (min)
C — Error propagation waterfall
(Layer 14 → 12 → 13)
Uncertainty (% of signal)
0.01%
3.00%
4.24%
4.90%
6.93%
3.46%
Electronic noise
Peak fit (σ_A/A)
Ratio (σ_R/R)
Normalise (σ_ρ/ρ)
Double ratio σ(Fk) (1 pair)
Avg 4 electrodes